

COM4513/COM6513: Mid-Semester Assignment

Building a Document Question Answering Assistant

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1) System Description

- **Chunking strategy:** Documents are tokenised and split into 400-token chunks with a 100-token overlap to prevent boundary loss. The top 6 chunks per question are retrieved using Term-frequency inverse document frequency (TF-IDF) and cosine similarity.
- **QA model:** deepset/deberta-v3-large-squad2
- **Answer selection:** Question tokens, special tokens and padding are masked so only passage tokens can be selected as answers. For each chunk, candidate start and end positions are paired to find the best valid span. Each span is then calibrated against the model's "no answer" confidence for that chunk.

2) Design Choices

Decision 1:

What: TF-IDF retrieval to select the top 6 chunks before model inference.

Why: The data set of choice was two 10-Ks and a 20-F which perfectly suit the implementation of TF-IDF because it down-weights terms that repeat excessively across boiler plate sections, it needs no training data and pairs naturally with cosine similarity for ranking chunks. Finally, once it has been built for a document it can be reused across every question at almost no extra cost.

Decision 2:

What: 100-token overlap between consecutive chunks.

Why: Without overlap, an answer spanning a chunk boundary would be truncated across both chunks. The overlap guarantees at least one chunk captures the full answer span with surrounding context.

3) Error Analysis

Incorrect Example 1:

- **Question:** Define "Owned Available Days"
- **Gold Answer:** the number of our ownership days less the aggregate number of days that our vessels are off-hire due to vessel familiarization upon acquisition, repairs, vessel upgrades or special surveys
- **Prediction:** 18,243 17,207
- **Why wrong:** This error highlights a limitation of TF-IDF: its reliance on lexical matching rather than semantic understanding leads to mathematical overweighting the exact n-gram "owned Available Days" the algorithm incorrectly prioritised a dense numerical table from the 10-K over textual definition.

Incorrect Example 2:

- **Question:** As of February 24, 2023, what was the size and total carrying capacity of the Company's fleet?
- **Gold Answer:** fleet of 44 drybulk vessels, with an aggregate carrying capacity of 4,455,600 dwt
- **Prediction:** 4,455,600 dwt
- **Why wrong:** The model selects a single continuous span, so compounding questions requiring two separate facts are reduced to whichever portion scores the highest. Here the numeric capacity figure had a stronger signal than the full fleet.

Improvement Suggestion: Although the TF-IDF chunk retrieval step performed well overall, replacing it with a semantic retrieval model- such as a sentence transformer- would most likely yield better results overall as it would better capture meaning-based relevance between questions and chunks.